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Modern Approaches For Automatic Question Paper Generator

[1] Gobinath Ramar
Christ University
Bangalore, Karnataka, India
gobinath.r@christuniversity.in

[2] Ramkumar Sivasakthivel
Christ University
Bangalore, Karnataka, India
ramkumar.s@christuniversity.in

[3] R. Stephen
Christ University
Bangalore, Karnataka, India
stephen.r@christuniversity.in

[4] R.V. Siva Balan
Christ University
Bangalore, Karnataka, India
sivabalan.rv@christuniversity.in

[5] Mayank Sahu
Christ University
Bangalore, Karnataka, India
mayank.sahu@bca.christuniversity.in

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***Abstract*— The Automatic question paper generation system in the educational field can be useful in improving the quality of question paper, distribution and evaluation process. The system can be used for maintaining the quality of the questions with higher accuracy and less error rate compared with other existing systems at a lower cost. This article gives a comprehensive literature survey of modern approaches followed in automatic question generation (AQPG) systems and categorizing the approaches used for the question paper generation process. The techniques such as rule-based, encoder-decoder based, generative adversarial network (GAN)-based, reinforcement learning-based, and transformer-based approaches are discussed in the paper and evaluated using standard metrics. The article presents insights into the strengths and limitations of each approach through the systematic comparison and analysis of multiple studies using BLEU-4, ROUGE-L, and METEOR metrics on the SQuAD dataset. The research finding of the article gives a better opportunity to the researcher and educators to improve the knowledge about automated question paper generator systems as well as the challenges incorporated during the implementation process of question paper generation. This article also gives a depiction of AI enabled solution in automated question paper generator.**

***Keywords*— Reinforcement Learning, Generative Adversarial Network, BLEU-4, ROUGE-L, METEOR, SQuAD.**

I. INTRODUCTION

In educational institutions the process of producing question papers is a repetitive and time consuming process for educators. The requirement to produce a large number of questions of good quality question papers without repetition is an extra burden on the teachers which leads to the reduction of the already scarce teaching time for the students. Despite the progress being made in the field of educational technology, most of the institutions still adopt the traditional approach to generate question papers, which is quite ineffective and often results in repetitive as well as low quality questions. These problems can be solved by Automatic Question Paper Generation (AQPG) systems that can help to reduce the time spent by educators on this task and increase the variety of questions. The field of AQPG has seen more advanced methods try to enhance question relevance and fluency through GANs and RL-based models. The most important advancement in AQPG research now focuses on transformer-based models which leverages T5, BERT and GPT architectures. These models have shown they can produce more varied and coherent questions by understanding contextual depths [1]. Syntactic rigidity along with decreased generalisability across subjects and dependency on predefined answer positions remain challenges for many prior approaches. The SQuAD dataset functions as a common benchmark to compare modern AQPG techniques through standard metrics like BLEU-4 [2], ROUGE-L [3], and METEOR [4] in the reviewed studies. This paper performs a five-category analysis of rule-based, encoder-decoder, GAN-based, RL-based and transformer-based models of AQG. The research aims to reveal both positive and negative aspects of different approaches with a special focus on the increasing application of transformer models in this field.

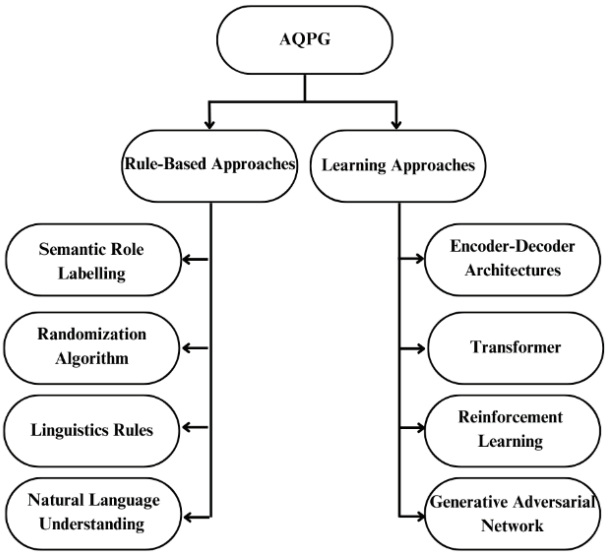


Fig. 1. Classification of Automatic question paper generator approaches.

II. LITERATURE REVIEW

A. Rule-based approaches

The paper has proposed a system for developing question papers automatically from a semantically annotated question bank. The paper involves creating a structured database of questions together with information like topic, difficulty level and cognitive skills like recall etc. It operated by first requiring users to input specifications such as topic weight, difficulty level and marks distribution before it applied semantic matching to select questions that matched the given requirements. This was confirmed by the testing where the user's requirements were followed and the papers were both diverse and appropriately difficult [5].

The authors recommend an adaptive question bank framework which retrieves questions from a structured question bank and generates question paper that match the

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criteria and the inputs provided by the Question Paper Designer (QPD). Model uses graph based concept mapping and store question in a database. It is also easy to implement and maintain however the system lacks scalability and question diversity as it is based on pre-existing limited question bank [6].

This paper introduces an rule-based approach for automatic generating questions from both simple and complex sentences. It uses NLTK POS Tagger, NLTK Parser, Stanford NER, and Porter Stemmer to preprocess the text, and then determine the clauses, and then apply question generation rules. The system also has the ability to identify verb phrases and some predefined rules to humanize "wh" (what, who, when, where, why, which, whose, and how) questions, yes/no questions, and descriptive questions. The evaluation was carried out with the help of two datasets: VerbNet and the QG 2010 Shared Task dataset. The framework used a rating scheme for testing, which showed an 80% agreement between human evaluators [7].

The paper shows an automatic question generation method which uses syntactic and semantic analysis through rule-based approaches. Using dependency-based, NER-based and semantic role labelling (SRL) the methodology transforms the sentences into questions. The patterns are applied with specific rules: S-V-ARGM-CAU (cause clause) and S-V-ARGM-TMP (temporal marker) which can choose to either produce rich or easy questions. Standard metrics like BLEU, METEOR, and ROUGE-L are used to evaluate the system across SQuAD datasets. The result shows that the proposed approach outperformed many comparable systems, with a higher improvement in METEOR scores, meaning that it is more effective at producing grammatically sound and meaningfully rich questions [8].

In this paper, the authors present a new way of automatic question generation (QG) task with the help of Natural Language Understanding (NLU) techniques. The main idea of their approach is the DeconStructure algorithm that first breaks down a sentence into its core elements via dependency parsing, semantic role labelling and PSG to get the meaning of the text better. The structured representation helps the system to produce more significant and relevant questions in the context. The model was compared with the state of the art QG systems and was found to have produced 71% more human evaluated acceptable questions. It also assigns a score to the questions according to their importance to the rest of the text and, as a result, enhances both question quality and diversity [9].

B. Encoder–decoder (sequence-to-sequence) architectures

Encoder and decoder architectures and long short term-memory (LSTM) networks for sequential tasks was presented by Google. Encoder–decoder models are typically used for modelling problems that are based on sequences as inputs and sequences as outputs. The encoder works on the input sequence as a fixed dimensional vector and the decoder outputs the target sequence. The present model is trained on paired input output data. The result shows that there is a great enhancement in machine translation and present the model's effectiveness in handling sequential tasks [10].

This paper describes an encoder-decoder architectures for automatic generation of questions from a human prompted text, using the SQuAD dataset. The author showcased how it can replaces Long Short Term Memory (LSTM) networks by GRU (Gated Recurrent Units) and it achieves the same accuracy, but with 15% less training time. It also shows the effectiveness of a convolutional encoder, which has evaluation scores very close to state of the art models, and reduces the training time by 25%. The results show the usefulness of the convolutional layers in capturing the core meaning of the text [11].

The authors discusses the paper about an Indonesian Automatic Question Generator based on sequence-to-sequence architecture with BiGRU, BiLSTM, and Transformer models. They have proposed to use linguistic features, copy mechanism, and coverage mechanism. Since there is no large public dataset for Indonesian AQG, the authors used SQuAD v2.0 in its original form for training and TyDiQA for testing. The model obtained BLEU-4 scores of 5.78 for SQuAD and 6.44 for TyDiQA. It showed better performance when answers were named entities that are close to their context. The study found that machine translation can introduce bias and unnatural patterns in data which can affects the performance of the model. However, the best cases produced acceptable and useful questions [12].

In this paper the authors presents an Graph Augmented Sequence-to-Sequence (GA-Seq2Seq) model for NQG(Neural Question Generation) which solves the issues present in the current Seq2Seq and Graph2Seq methods. The model comprises of three main components: (1) An answer-aware passage representation module that combines answer information, (2) A graph-augmented passage encoder that integrates both graph and sequence encoders to extract both structural and semantic information and, (3) An attention-based LSTM decoder that produces fluent questions. GA-Seq2Seq achieved superior performance over baselines on both the SQuAD and MS MARCO datasets through automatic and human evaluations. GA-Seq2Seq produced more coherent and relevant questions, showing strong potential as an approach for question generation applications in reading comprehension, question answering, and dialogue systems [13]

The conceptualized model for automatic question generation is a dual attention based encoder decoder model. Bidirectional LSTMs are used for both answer and paragraph encoding to capture contextual dependencies. The attention mechanism helps to align the input text with the generated question and the answer with the input, this helps the model to improve relevance. A two layer LSTM is used as decoder to generate fluent questions which integrates attention based context vectors. In Keras with TensorFlow, the model was implemented with a source vocabulary of 50000 and a target vocabulary of 35000, Nadam optimizer (LR = 0.001), sparse cross entropy loss, and beam search (width = 5) for decoding. 10 epochs of training with a batch size of 32 was used to make sure that question generation was both efficient and accurate [14].

C. Generative adversarial network

GANs was introduced as a novel method for training generative models via competition in the paper. It is a two-part system: the generator (G), it is able to produce data of the same distribution as the training data and the discriminator (D), it tries to distinguish between real data and data produced by the generator. The two networks are learned simultaneously in a minimax model, G is trying to convince D that the produced data is real while D is trying not to be convinced. Both models are used as multilayer perceptron and use backpropagation for

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training so no Markov chains or approximate inference is needed. It was shown in experiments that GANs are able to produce good quality samples, with G learning the data distribution and D converging to 50% accuracy which means that the outputs are indistinguishable. This framework appeared to be potentially very useful for deep generative modelling [15]

The paper explores a joint learning approach for Question Answering (QA) and Question Generation (QG) which can enhance both the tasks by taking into account their connection. It introduces two training algorithms: (1) an extended Generative Adversarial Network (GAN) that chooses to incorporate the generated questions into the QA training and (2) an extended dual learning framework that shows the relationship between the two tasks as a regularization feature. For document and table-based QA tasks, the experiments were performed to see whether both algorithms enhance the accuracy of QA and BLEU scores of QG. It was observed in the experiment that policy gradient helps to improve the quality of QG through its connection with QA, whereas applying GAN directly to treat all the generated questions as the negative instances does not necessarily enhance the QA. The adaptive collaborative learning strategy provides the best performance boost as compared to the purely competitive one. Therefore, the results show that joint learning with selective collaboration can greatly improve both QA and QG models respectively [16].

This paper proposed a neural network model for question generation, which uses a latent variable (z) to ensure the diversity of questions and an observed variable(c) to regulate question type. The model is implemented in the Generative Adversarial Network (GAN) architecture, and the discriminator not only has the ability to distinguish whether an input is a real question or not, but also to predict the type of the question. The model was trained and tested on the SQuAD dataset. The results show that the "seq2seq + z + c + GAN" approach outperforms other models in terms of diversity and readability of the generated questions. The latent variable alone captures the diversity, while the observed variable enhances the specificity of the model. The performance is also improved by the GAN-based adversarial learning. Upon human evaluation it can be said that the questions generated are natural and relevant to the context more than the previous models [17].

The paper tries to solve the problem of the absence of labelled data, DoubAN (Doubly Adversarial Nets) is proposed to improve automatic question generation (QG) in one domain through the use of a labelled dataset from another domain. The model is broken down into two main components: the question generator and two adversarial components, namely, the Domain-Classification Discriminator (DC-Dis) to assist the generator to learn domain general representations and the Question Answering Discriminator (QA-Dis) to provide an estimated reward which would help in enhancing the quality of questions generated. With the SQuAD dataset used as the target domain and the NewsQA dataset used as the source domain, the experiments show that DoubAN outperforms the existing baseline models in terms of BLEU-4 scores. The results also showed that both the adversarial components are interacting indirectly and both of them are enhancing the question generation in their own way and the full model is the best. Furthermore, this study talks about the challenges in domain differences and evaluation metrics and pointed out that using BLEU scores alone may not always be enough to capture the grammaticality and the relevance of the questions being generated [18].

D. Reinforcement Learning

The authors propose a framework to overcome prior method limitations which neglect structural information and use only cross-entropy loss or fail full answer integration. In this paper they develop a reinforcement learning (RL) based Graph-to-Sequence (Graph2Seq) model for natural question generation (QG). The model includes a Bidirectional Gated Graph Neural Network (GNN) to process the passage as a graph and a Deep Alignment Network (DAN) to include the answer information at both word and contextual levels. A hybrid evaluator combines both cross-entropy loss and RL goals to optimize question generation output while still ensuring syntactic and semantic questions are generated. The model was tested on SQuAD dataset, where it significantly outperformed existing methods [19].

The paper demonstrates a new framework to perform automatic question generation tasks (QG) through deep reinforcement learning (RL). The proposed approach differs from traditional sequence-to-sequence (Seq2Seq) models as the traditional methods depend on maximum likelihood estimation (MLE) and experience exposure bias because they allow an evaluator model to supply reinforcement signals for better question quality. This generator uses Sequence-to-Sequence architecture with pointer networks along with copy and coverage mechanisms to improve both answer prediction and context preservation as well as also maintaining word diversity. The evaluation uses BLEU-4 ROUGE-L and three QG-specific metrics known as Question Syntactic Score (QSS) and Answerability Score (ANSS). The model showed superior performance because it uses reinforcement learning for optimization instead of MLE-based training to align training and evaluation objectives [20].

In this paper the authors presented an Answer-Driven Deep Question Generation (ADDQG) as a new method which uses answer information directly to produce questions that both concerns to the text context to exhibit improved contextual complexity. ADDQG differs from traditional approaches because they only modified document representations and it applies an answer-aware initialization module to balance document and answer information via a gated connection layer. It uses a semantic-rich fusion attention mechanism uses Graph Attention Networks (GATs) to improve multi-hop reasoning abilities which results in better deep question generation. The model achieves improved results through reinforcement learning (RL) training with two rewards for syntactic accuracy (BLEU-4) and semantic relevance (Word Mover's Distance), this helps to reduce exposure bias. The combination leads to simple, but precise questions which surpass benchmark models on the HotpotQA dataset with better results in BLEU-4, METEOR, and ROUGE-L scores [21].

This paper focuses on reinforcement learning for the optimization of question generation (QG) systems by considering fluency, relevance, and the ability of a question to be answered. The authors model is different from traditional sequence-to-sequence models that are trained with maximum likelihood estimation, the authors come up with three task-specific rewards which are; (1) Fluency which is measured by the perplexity of a pre-trained language model; (2) Relevance which is measured by BERT classifier that differentiates

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between high quality questions; and (3) Answerability which is evaluated via a SpanBERT-based question answering model to determine whether the generated question has a real answer. The rewards were optimized using reinforcement learning with baseline reward approach to stabilize training. Another difference is that they thoroughly investigated the relation of reward types to human judgment. They conclude that relevance rewards greatly enhance the quality of questions, however, answerability rewards bring biases into the picture which indicates that rewards for QG should match well with human evaluation criteria [22].

E. Transformer

The Transformer is a novel sequence transformation model which removes the need of recurrent and convolutional architectures because it uses a self-attention mechanisms and positional encodings instead. The self-attention mechanism enables the model to assign weights to the words in a sequence so that they relate to one another this in turn results in better contextual comprehension awareness. The Transformer uses multi-head attention that operates via multiple attention mechanisms in parallel to detect a range of linguistic features. This model achieves significant speedup in parallelization leading to a faster training process when compared to traditional recurrent neural networks (RNNs) and convolutional neural networks (CNNs). The Transformer attains a BLEU score of 28.4 on the WMT 2014 English-to-German translation task and surpasses previous models including ensembles. It achieves a BLEU score of 41.8 on the WMT 2014 English-to-French task and requires 3.5 days of training when using 8 GPUs, which can be considered a significant improvement in efficiency [23].

This paper presents a T5-based text-to-text transformer model for the task of generating question answer pairs from text. The model produces different question types such as subjective (short and long answer), Boolean (yes/no), fill in the blank, and multiple-choice questions to guarantee the variety and flexibility of the assessment tool. Google Colab is used for training the model with PyTorch environment together they are employed with Hugging Face's Transformers library. The model was benchmarked on SQuAD, QuAC, and BoolQ datasets with grid search being used to tune hyperparameters like learning rate (1e-4), batch size (32), and training epochs (18). Sequence decoding was performed using beam search with a size of 4 to improve question quality. The model achieved better results than the baseline approaches on both BLEU-4 (18.87) and METEOR (25.24) metrics that show that the model produced grammatically correct, contextually appropriate, and diverse question answer pairs. The system can efficiently handle both sentence and paragraph levels of context while earlier models struggled with context-based question formation as reported in [24].

The paper explores simplifying paragraph-level question generation using GPT-2, a transformer-based model. In addition, the authors tuned the GPT-2 model with causal language modelling loss for several causal language datasets such as SQuAD. They used an Adam optimizer (learning rate: 5e-4) for 3 epochs and Tesla V100 GPU. They evaluated through different data formatting (One Question Per Line vs. All Questions Per Line) and varied output was reached through top-p nucleus sampling ($p = 0.9$). The best model in the experiment was OQPL with number delimiters, and it performed better than previous RNN-based models in METEOR and ROUGE-L. However, the model over-relied on context copying for generating identification type questions like who/what/when/where, which may be due to dataset bias [25].

This paper presents a new framework for automatic question generation (QG) which combines GPT-2 as a transformer-based question generator and BERT as question answering (QA) model. The system is trained in an end-to-end fashion where the QG model generates questions given the context and the QA model provides feedback on those questions. First, BERT is pre-trained for QA and then GPT-2 is fine-tuned for QG. They evaluated the system on SQuAD dataset using conventional metrics such as BLEU and ROUGE-L along with a new replacement metric that was tied to the QA performance. The outcomes show that incorporating BERT's bi-directional contextual embeddings as reusable diagnostics improves the quality of QG, especially in semi-supervised settings, which approached near-supervised performance with minimal labelled data [26].

The study describes an automated question-answer generation system which is based on a text-to-text transfer transformer (T5) model. The methodology is based on a stacked encoder-deocoder architecture which had been tuned for question generation and answer extraction. The system can take input text, determine the key sentences, extract the answers and then produce the corresponding questions in different formats such as subjective, multiple choice, boolean and fill in the blank. The model was trained with datasets such as SQuAD, QuAC, BoolQ and evaluated with BLEU-4, ROUGE-L, METEOR, and F1 metrics. The results showed that the model outperforms the existing models in terms of performance. The study also concluded that the system is able to produce efficient, diverse, contextually accurate and grammatically correct question answer pairs which can be used for educational assessments and self-learning [27].

In this paper, the authors present a new way of generating question–answer pairs with difficulty control based on Item Response Theory (IRT). While most of the current methods do not take into account the ability of learners and therefore fail to quantify the difficulty of the questions, this approach adapts the questions to the learners using IRT. The methodology modifies the pretrained transformer models such as BERT and T5 by incorporating difficulty values to enable the adaptive question generation. Furthermore, the system utilizes computerized adaptive testing (CAT) to dynamically estimate the ability of the learner and then modify the difficulty of the questions accordingly. It is a dual approach of personalization to the learning experiences as well as effective question generation based on the ability of each learner [28].

The paper proposes a framework by demonstrating a new way to humanize conversational questions by combining topic modelling with transformer-enabled models and applying genetic algorithms to optimize the process. This method shows improved relevance by bringing topic awareness into the question generation process which helps maintain question relevance to the conversation. The framework uses T5 transformer model and fine-tuned its weights using genetic algorithm. BERT combined with c-TF-IDF (Class-based-term frequency-inverse document frequency) produces topic context from past conversation history which helps generate dense clusters of related topics. The model was trained with the CoQA dataset before qualitative evaluation [29]

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III. Evaluation

A comparative assessment of different Automatic Question Paper Generation (AQPG) approaches based on the approaches taken, training dataset employed, and performance measures is presented in Table I. Rule-based approaches are dependent on pre-defined linguistic rules, semantic role labelling (SRL) and natural language understanding (NLU) which guarantee a structured approach to question creation but fail to provide variety and flexibility. Encoder-decoder models like Seq2Seq and graph augmented Seq2Seq enhance the contextual relevance using neural networks and obtain better BLEU, METEOR and ROUGE scores than the rule-based systems but they show inconsistencies in the performance of GANs which employ adversarial learning for diversity. Reinforcement learning (RL) based methods learn the question generation process iteratively to improve the quality of the questions and therefore the coherent and fluent questions are produced. Among the approaches, transformer-based models especially T5 and GPT-2 are found to perform best in terms of evaluation metrics, producing fluent, appropriate, and diverse questions. These findings support the recent trend of using transformer-based models in state-of-the-art AQG systems.

Research papers based on their key focus areas are categorized in Table II. The table reveals that the focus of the primary research in AQPG is on fluency, accuracy and the contextual relevance of the generated questions. Most of the models are aimed at the generation of syntactically and semantically correct questions, and still be contextually relevant to the input text. Question variability is the next focus point, particularly in approaches based on Generative Adversarial Networks (GANs) and Reinforcement Learning

TABLE I. Comparison of Question Generation Methods Based on Technique, Dataset, and Metrics.

Methodology	Reference	Technique	Dataset	Metrics			Description
				BLEU-4	*METEOR*	*ROUGE-L*	
Rule-based approaches	[5]	Rule-Based	STQR	–	–	–	Utilizes semantically tagged questions for user-specifications
	[6]	Rule-Based	Question bank	–	–	–	Uses predefined rules to manage and generate adaptive questions.
	[7]	Rule-Based	VerbNet	–	–	–	Generates questions from simple and complex sentences using linguistic rules.
	[8]	Semantic Role Labelling	SQuAD	10.61	25.01	40.38	Uses SRL and dependency parsing to generate semantically rich questions.
	[9]	Natural Language Understanding (NLU)	SQuAD	–	–	–	Integrates NLU techniques to improve question generation accuracy.
Encoder-decoder	[12]	Seq2Seq	SQuAD	5.78	–	43.4	Uses Seq2Seq with attention for generating Indonesian questions.
	[13]	Graph-Augmented Seq2Seq	SQuAD	18.96	22.52	47.14	Uses graph-based encoder to enhance question generation.
			MS MARCO	21.79	26.91	51.66	
	[14]	Dual Encoder with Neural Attention	SQuAD	18.27	23.08	42.97	Uses dual encoders and attention to improve question generation.
Generative adversarial network	[16]	Joint Learning (QA + QG)	MS MARCO	17.61	–	–	Trains Question Answering and Question Generation models together to enhance performance.
	[17]	Doubly Adversarial Nets	SQuAD	13.36	17.7	40.42	Uses two adversaries—domain classifier and QA model—for QG.
	[18]	Doubly Adversarial Nets	SQuAD	5.48	–	–	Uses two adversaries—domain classifier and QA model—for QG.
Reinforcement Learning	[19]	RL + Graph-to-Sequence	SQuAD	18.12	21.73	46	Uses a graph-structured encoder with RL to improve QG.
	[20]	Deep RL	SQuAD	16.48	20.21	44.11	Trains QG using RL-based policy gradient optimization.
	[21]	Answer-Driven RL	HotpotQA	17.54	20.56	38.09	Uses answer-aware RL-based QG for improving coherence.
	[22]	Question-Specific Reward RL	HotpotQA	15.41	19.61	35.12	Introduces question-specific rewards for diverse and deep QG
Transformer	[24]	Text-to-Text Transfer Transformer	SQuAD	18.87	25.24	40.64	Uses T5 to generate diverse and grammatically correct QG pairs.
	[25]	GPT-2 Finetuning	SQuAD	8.27	21.2	44.41	Uses GPT-2 with fine-tuning for paragraph-level QG.
	[26]	GPT-2 + BERT Hybrid Model	SQuAD	7.84	–	34.51	Merges GPT-2 (QG) and BERT (QA) for bidirectional learning.
	[27]	Transformer-based Seq2Seq Model	Not specified	17.59	28.03	49.66	Leverages transformer models to map sentences from text passages directly to questions.

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TABLE II. CLASSIFICATION OF RESEARCH PAPERS BY KEY FOCUS AREAS

Reference	Fluency & Accuracy	Contextual Relevance	Question Variability	Computational Efficiency	Expert & User Assessment
G. Nalawade & R. Ramesh	–	–	–	✓	–
V. K. Purohit et al.	–	–	–	✓	–
R. Das et al.	✓	–	✓	–	–
O. Keklik et al.	✓	✓	–	–	–
K. Mazidi & P. Tarau	–	✓	–	–	✓
F. J. Muis & A. Purwarianti	✓	–	–	✓	–
H. Ma et al.	✓	✓	–	–	–
A. P. Kumar et al.	✓	✓	–	–	–
Y. Sun et al.	✓	✓	–	–	–
K. Yao et al.	–	✓	–	–	✓
J. Bao et al.	–	–	✓	✓	–
Y. Chen et al.	✓	✓	–	–	–
V. Kumar et al.	✓	✓	–	–	–
L. Wang et al.	–	✓	✓	–	–
Y. Xie et al.	✓	–	✓	–	–
R. Goyal et al.	✓	–	✓	–	–
L. E. Lopez et al.	–	✓	✓	–	–
T. Klein & M. Nabi	–	✓	–	✓	–
A. Barlybayev & B. Matkarimov	✓	✓	–	–	–

(RL), which aim to diversify the generated questions. Only a few papers focus on computational efficiency, by optimizing the processing time or using efficient architectures like Transformers. Expert & user assessment is the least explored focus area. The trend in the AQPG research can be seen as the gradual advancement towards Transformer models which achieve consistently higher evaluation scores.

IV. CONCLUSION

The automatic question paper generator followed in recent years faces various issues in predicting exact combinations of questions according to the user specifications. The modern tools available for predicting the questions are not accurate and takes much time in generating the questions. Some of the automated tools gives irrelevant information, which creates huge understandability problems for the educator or the exam instructor. The examinations carried out in this research article gives a detailed review of methods, techniques implemented in existing automated question paper generator.

This article examines the existing work carried out in rule-based encoder, reinforcement learning, generative adversarial networks and transformer-based approaches with necessary metrics. The metrics such as BLEU-4, ROUGE-L and METEOR, which are widely used in existing methodologies are used to compare the existing automatic question paper generator systems.

The analysis revealed that the most frequent focus was on fluency and accuracy, which means ensuring the syntactic and semantic correctness of the questions, although the contextual relevance still played an important role in the process of matching question relevance with source texts. Adversarial learning and reinforcement learning models were particularly emphasized for their ability to increase the diversity of the questions generated in terms of question variability. Computational efficiency was prioritized in studies optimizing for processing time and architectural design.

Human evaluation was the least common metric, indicating a gap in user-based validation. Some of the potential areas for future work are context-aware question generation and human in the loop evaluation to enhance the accuracy and naturalness of the questions.

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REFERENCES

[1] N. Mulla and P. Gharpure, “Automatic question generation: a review of methodologies, datasets, evaluation metrics, and applications,” Progress in Artificial Intelligence, Jan. 2023, doi: https://doi.org/10.1007/s13748-023-00295-9.

[2] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, “BLEU: a Method for Automatic Evaluation of Machine Translation,” Proceedings of the 40th Annual Meeting on Association for Computational Linguistics - ACL ’02, 2001, doi: https://doi.org/10.3115/1073083.1073135.

[3] S. Banerjee and A. Lavie, “METEOR: An Automatic Metric for MT Evaluation with Improved Correlation with Human Judgments,” ACL Anthology, pp. 65–72, Jun. 2005.

[4] C.-Y. Lin, “ROUGE: A Package for Automatic Evaluation of Summaries,” aclanthology.org, Jul. 01, 2004.

[5] G. Nalawade and R. Ramesh, "Automatic Generation of Question Paper from User Entered Specifications Using a Semantically Tagged Question Repository," 2016 IEEE Eighth International Conference on Technology for Education (T4E), Mumbai, India, 2016, pp. 148-151, doi: https://doi.org/10.1109/T4E.2016.038.

[6] V. K. Purohit et al., “Design of adaptive question bank development and management system,” pp. 256–261, Dec. 2012, doi: https://doi.org/10.1109/pdgc.2012.6449828.

Authorized licensed use limited to: Christ University. Downloaded on September 23,2025 at 04:41:44 UTC from IEEE Xplore. Restrictions apply.

[7] R. Das, A. Ray, S. Mondal and D. Das, "A rule based question generation framework to deal with simple and complex sentences," 2016 International Conference on Advances in Computing, Communications and Informatics (ICACCI), Jaipur, India, 2016, pp. 542-548, doi: https://doi.org/10.1109/ICACCI.2016.7732102.

[8] O. Keklik, T. Tuglular, and S. Tekir, "Rule-Based Automatic Question Generation Using Semantic Role Labeling," IEICE Transactions on Information and Systems, vol. E102.D, no. 7, pp. 1362–1373, Jul. 2019, doi: https://doi.org/10.1587/transinf.2018edp7199.

[9] K. Mazidi and P. Tarau, "Infusing NLU into Automatic Question Generation," University of North Texas Digital Library (University of North Texas), pp. 51–60, Jan. 2016, doi: https://doi.org/10.18653/v1/w16-6609.

[10] I. Sutskever, O. Vinyals, and Q. V. Le, "Sequence to Sequence Learning with Neural Networks," arXiv.org, 2014. https://arxiv.org/abs/1409.3215

[11] J. Singh and Y. Sharma, "Encoder-Decoder Architectures for Generating Questions," Procedia Computer Science, vol. 132, pp. 1041–1048, 2018, doi: https://doi.org/10.1016/j.procs.2018.05.019.

[12] F. J. Muis and A. Purwarianti, "Sequence-to-Sequence Learning for Indonesian Automatic Question Generator," 2020 7th International Conference on Advance Informatics: Concepts, Theory and Applications (ICAICTA), Tokoname, Japan, 2020, pp. 1-6, doi: https://doi.org/10.1109/ICAICTA49861.2020.9429032.

[13] H. Ma, J. Wang, H. Lin, and B. Xu, "Graph augmented sequence-to-sequence model for neural question generation," Applied Intelligence, vol. 53, no. 11, pp. 14628–14644, Oct. 2022, doi: https://doi.org/10.1007/s10489-022-04260-2.

[14] A. P. Kumar, G. Sridhar, A. Nayak, and M. K. Shenoy, "A Neural Attention Model for Automatic Question Generation Using Dual Encoders," Advances in intelligent systems and computing, pp. 427–440, Oct. 2021, doi: https://doi.org/10.1007/978-981-16-3342-3_34.

[15] I. Goodfellow et al., "Generative adversarial networks," Communications of the ACM, vol. 63, no. 11, pp. 139–144, Oct. 2020, doi: https://doi.org/10.1145/3422622.

[16] Y. Sun et al., "Joint Learning of Question Answering and Question Generation," in IEEE Transactions on Knowledge and Data Engineering, vol. 32, no. 5, pp. 971-982, 1 May 2020, doi: https://doi.org/10.1109/TKDE.2019.2897773

[17] K. Yao, L. Zhang, T. Luo, L. Tao, and Y. Wu, "Teaching Machines to Ask Questions," Ijcai.org, pp. 4546–4552, 2018, doi: https://doi.org/10.24963/ijcai.2018/632.

[18] J. Bao, Y. Gong, N. Duan, M. Zhou and T. Zhao, "Question Generation With Doubly Adversarial Nets," in IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 26, no. 11, pp. 2230-2239, Nov. 2018, doi: https://doi.org/10.1109/TASLP.2018.2859777.

[19] Y. Chen, L. Wu, and M. J. Zaki, "Reinforcement Learning Based Graph-To-Sequence Model for Natural Question Generation," International Conference on Learning Representations (ICLR) 2020. pp. 498–515, doi: https://doi.org/10.48550/arXiv.1908.04942

[20] V. Kumar, G. Ramakrishnan, and Y.-F. Li, "A framework for automatic question generation from text using deep reinforcement learning," Aug. 15, 2018 doi:https://dx.doi.org/10.48550/arXiv.1808.04961.

[21] L. Wang, Z. Xu, Z. Lin, H. Zheng, and Y. Shen, "Answer-driven Deep Question Generation based on Reinforcement Learning," Proceedings of the 17th international conference on Computational linguistics -, Jan. 2020, doi: https://doi.org/10.18653/v1/2020.coling-main.452.

[22] Y. Xie, L. Pan, D. Wang, M.-Y. Kan, and Y. Feng, "Exploring Question-Specific Rewards for Generating Deep Questions," Proceedings of the 28th International Conference on Computational Linguistics, pp. 2534–2546, Jan. 2020, doi: https://doi.org/10.18653/v1/2020.coling-main.228.

[23] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," Advances in Neural Information Processing Systems, vol. 30, pp. 5999–6009, 2017, doi: https://doi.org/10.48550/arXiv.1706.03762

[24] R. Goyal, P. Kumar, and V. P. Singh, "Automated Question and Answer Generation from Texts using Text-to-Text Transformers," Arabian journal for science and engineering, vol. 49, no. 3, pp. 3027–3041, May 2023, doi: https://doi.org/10.1007/s13369-023-07840-7.

[25] L. E. Lopez, D. K. Cruz, J. Christian, and C. Cheng, "Simplifying Paragraph-Level Question Generation via Transformer Language Models," Lecture notes in computer science, pp. 323–334, Jan. 2021, doi: https://doi.org/10.1007/978-3-030-89363-7_25.

[26] T. Klein and M. Nabi, "Learning to Answer by Learning to Ask: Getting the Best of GPT-2 and BERT Worlds," arXiv:1911.02365 [cs], Nov. 2019, doi: https://doi.org/10.48550/arXiv.1911.02365

[27] Alibek Barlybayev and Bakhyt Matkarimov, "Development of system for generating questions, answers, distractors using transformers," International Journal of Power Electronics and Drive Systems/International Journal of Electrical and Computer Engineering, vol. 14, no. 2, pp. 1851–1851, Jan. 2024, doi: https://doi.org/10.11591/ijece.v14i2.pp1851-1863.

[28] Y. Tomikawa, A. Suzuki and M. Uto, "Adaptive Question–Answer Generation With Difficulty Control Using Item Response Theory and Pretrained Transformer Models," in IEEE Transactions on Learning Technologies, vol. 17, pp. 2186-2198, 2024, doi: https://doi.org/10.1109/TLT.2024.3491801

[29] N. Mulla and P. Gharpure, "Genetic Algorithm Optimized Topic-aware Transformer-Based Framework for Conversational Question Generation," Procedia Computer Science, vol. 230, pp. 914–922, 2023, doi: https://doi.org/10.1016/j.procs.2023.12.041.

[30] P. Rajpurkar, J. Zhang, K. Lopyrev, and P. Liang, "SQuAD: 100,000+ Questions for Machine Comprehension of Text," Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP 2016), Austin, TX, USA, Nov. 2016, pp. 2383–2392, doi: https://doi.org/10.18653/v1/D16-1264.

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